THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 6,04,800	1 day = 86,400 seconds.
2	**(c)** 15	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
3	**(d)** 0.024	Multiply two at a time and count the decimal places carefully.
4	**(b)** 21	Square first, then multiply.
5	**(a)** 90°	Reason with the angle relations.
6	**(c)** 108	Find the first multiple of 9.
7	**(c)** 3	List the triples $a \le b \le c$ with $a + b > c$.
8	**(d)** $\frac{1}{4}$	Work the bracket first.
9	**(c)** 7	Count zeros in the product.
10	**(d)** 15	Sum first, then double, then subtract.
11	**(c)** 48	Divide the larger number by the smaller.
12	**(d)** $2x + 13$	Expand each bracket, watching the signs.
13	**(b)** 120°	Share 180° in the given ratio.
14	**(d)** 9873	Start from 9876 and test downwards.
15	**(d)** 40°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
16	**(b)** 135	Find the number first.
17	**(d)** 25	Divide by 1,00,00,000.
18	**(c)** -4	Evaluate both products.
19	**(c)** 7	Divide the part by the fraction it represents.
20	**(a)** $16x + 6$	Perimeter $= 2\times$(length + breadth).
21	**(c)** 110°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
22	**(b)** 5	Repeat the routine and count.
23	**(c)** 90°	Reason with the angle sum and exterior angle property.
24	**(b)** 30	Time for the whole = time for the part ÷ fraction.
25	**(b)** 48	Divide.
26	**(c)** 2277	Distributive property.
27	**(b)** 1.7	Work out the bracket first.
28	**(b)** 55	Substitute, multiply, then halve.
29	**(b)** 290°	A complete turn is 360°.
30	**(b)** 9	Follow the rule term by term.
31	**(b)** 80°	Find x from the angle sum, then compare the angles.
32	**(a)** $\frac{1}{3}$	Unpainted = (unpainted before) $\times$ (what is not painted of it).
33	**(a)** 43,211	Round up to the next lakh.
34	**(b)** 100	Distributive property.
35	**(c)** 0.95	Find how far each number is from 1.
36	**(a)** $2n + 4$	Test the expressions with small values of n.
37	**(c)** 55°	Larger + smaller $= 90^\circ$ and larger − smaller $= d$.
38	**(a)** even	Test with small values.
39	**(c)** no triangle can be formed	Check the sum of the two shorter sides.
40	**(b)** greater than 1	Reason about the size of products and quotients.
41	**(a)** Both A and R are true and R is the correct explanation of A.	1,00,00,000 / 1,00,000 = 100.

42	**(a)** Both A and R are true and R is the correct explanation of A.	3 x 4 = 12 and 5 + 12 = 17.
43	**(a)** Both A and R are true and R is the correct explanation of A.	0.50 > 0.45.
44	**(a)** Both A and R are true and R is the correct explanation of A.	Three times n is 3n, and five more is 3n + 5.
45	**(a)** Both A and R are true and R is the correct explanation of A.	90 - 35 = 55.
46	**(a)** Both A and R are true and R is the correct explanation of A.	2m + 1 + 2n + 1 = 2(m + n + 1), which is even.
47	**(a)** Both A and R are true and R is the correct explanation of A.	3 + 4 = 7 < 8, so the rule fails.
48	**(a)** Both A and R are true and R is the correct explanation of A.	6/12 = 1/2.
49	**(b)** Both A and R are true but R is not the correct explanation of A.	1,000,000 = 10,00,000 = 10 lakh, and 1 billion = 1000 million. Both true, but R does not explain A.
50	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain A.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** $\frac{1}{8}$	Write over 1000, 100 or 10 and reduce.
2	**(b)** 23	Apply the rule to the input.
3	**(b)** 120°	Total $= 180^\circ$ in 3 equal parts.
4	**(b)** 3	Add the reverse each time.
5	**(b)** 30°	They add up to 90°.
6	**(c)** 1	Find x first.
7	**(c)** 1800	1 crore = 100 lakh.
8	**(b)** 24	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
9	**(a)** $\frac{1}{2}$	Convert every number to a decimal and compare.
10	**(c)** 77	$x^2 = x \times x$.
11	**(a)** parallel	Reason with the angle relations.
12	**(c)** $(10a + b) - (10b + a)$	Difference of a number and its reverse is $9(a - b)$.
13	**(d)** 3	List a, a, b with $b = P - 2a$ and $b < 2a$.
14	**(d)** $\frac{1}{3}$	Work the bracket first.
15	**(a)** 25,92,000	1 day = 86,400 seconds.
16	**(b)** 82	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
17	**(c)** 0.18	Multiply two at a time and count the decimal places carefully.
18	**(c)** 13	Square first, then multiply.
19	**(c)** 108°	Share 180° in the given ratio.
20	**(c)** 1008	Find the first multiple of 9.
21	**(a)** 3	List the triples $a \le b \le c$ with $a + b > c$.
22	**(b)** 84	Find the number first.
23	**(c)** 7	Count zeros in the product.
24	**(b)** 25	Sum first, then double, then subtract.
25	**(b)** 24	Divide the larger number by the smaller.
26	**(a)** $3x + 29$	Expand each bracket, watching the signs.
27	**(d)** 125°	Complement $= 90^\circ - a$; supplement of that $= 180^\circ - (90^\circ - a)$.
28	**(a)** 9875	Start from 9876 and test downwards.
29	**(d)** 70°	Base angle $= 180^\circ - e$; apex $= 180^\circ - 2\times$ base.
30	**(a)** 20	Time for the whole = time for the part ÷ fraction.
31	**(d)** 4.5	Divide by 1,00,00,000.
32	**(a)** -6	Evaluate both products.
33	**(c)** 8	Divide the part by the fraction it represents.
34	**(d)** $18x + 6$	Perimeter $= 2\times$(length + breadth).
35	**(a)** 225°	A complete turn is 360°.
36	**(b)** 7	Repeat the routine and count.
37	**(b)** 360°	Reason with the angle sum and exterior angle property.
38	**(a)** $\frac{1}{5}$	Unpainted = (unpainted before) × (what is not painted of it).
39	**(b)** 36	Divide.
40	**(b)** 4455	Distributive property.
41	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; a zero at the end of a decimal part does not change the value, but R does not explain it.

42	**(b)** Both A and R are true but R is not the correct explanation of A.	2 x 4 + 3 = 11. Both are true, but R does not explain the value.
43	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but equality of vertically opposite angles does not explain the linear pair.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	5 + 8 + 3 + 2 = 18 is divisible by 9, so A is true. R is true but does not explain A.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Two right angles would make 180 degrees with nothing left for the third angle. Both true, but R does not explain A.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	(3/4) x 2 = 3/2 and (1/2)(3/4) = 3/8. Both true but separate facts.
47	**(c)** A is true but R is false.	A is true. The international system puts a comma after every three digits, so R is false.
48	**(c)** A is true but R is false.	(20 - 8) - 4 = 8, so A is true. 20 - (8 - 4) = 16, so R is false.
49	**(c)** A is true but R is false.	The digit after the hundredths place (5, 6) decides: 3.456 rounds to 3.46, so A is true. 6 is in the thousandths place, so R is false.
50	**(c)** A is true but R is false.	Like terms are added by adding their numbers: 5a. The letter part does not change, so R is false.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–8) | Class 7 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 70°	Larger + smaller = 180°.
2	**(c)** 3	Test with small values.
3	**(d)** a triangle can be formed	Check the sum of the two shorter sides.
4	**(b)** 2	Multiply the fraction by the mixed number.
5	**(b)** 340	Divide by a lakh.
6	**(b)** 94	Add the decreasing distances.
7	**(c)** 1.125	Subtract the smaller number from the larger.
8	**(d)** 9	Substitute carefully; xy means $x \times y$.
9	**(b)** 135°	Total = 180° in 4 equal parts.
10	**(b)** 28	Triangular numbers: $n(n+1)/2$.
11	**(c)** 14	Base = $P - 2s$.
12	**(c)** 1	Find x first.
13	**(d)** 9	Multiply and count.
14	**(c)** 7	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
15	**(c)** $\frac{3}{8}$	Write over 1000, 100 or 10 and reduce.
16	**(d)** 29	Apply the rule to the input.
17	**(c)** 90°	Reason with the angle relations.
18	**(d)** 6	Add the reverse each time.
19	**(c)** 36°	They add up to 90°.
20	**(a)** $\frac{1}{3}$	Work the bracket first.
21	**(b)** 3500	1 crore = 100 lakh.
22	**(c)** 21	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
23	**(d)** $\frac{7}{8}$	Convert every number to a decimal and compare.
24	**(c)** 79	$x^2 = x \times x$.
25	**(b)** 100°	Share 180° in the given ratio.
26	**(b)** $(10a + b) + (10b + a)$	The sum is $11(a + b)$.
27	**(b)** 4	List a, a, b with $b = P - 2a$ and $b < 2a$.
28	**(d)** 135	Find the number first.
29	**(b)** 86,40,000	1 day = 86,400 seconds.
30	**(b)** 32	Apply BODMAS: brackets, orders, division/multiplication, addition/subtraction.
31	**(b)** 0.2	Multiply two at a time and count the decimal places carefully.
32	**(d)** 32	Square first, then multiply.
33	**(a)** 140°	Complement = $90^\circ - a$; supplement of that = $180^\circ - (90^\circ - a)$.
34	**(c)** 10008	Find the first multiple of 9.
35	**(d)** 4	List the triples $a \le b \le c$ with $a + b > c$.
36	**(d)** 24	Time for the whole = time for the part ÷ fraction.
37	**(d)** 6	Count zeros in the product.
38	**(b)** 27	Sum first, then double, then subtract.
39	**(c)** 30	Divide the larger number by the smaller.
40	**(c)** $4x + 14$	Expand each bracket, watching the signs.
41	**(c)** A is true but R is false.	A is true. Vertically opposite angles are equal, they do not always add up to 180 degrees, so R is false.

42	**(c)** A is true but R is false.	The sum is 11(a + b), so A is true; R is false because the correct result is 11(a + b).
43	**(c)** A is true but R is false.	A is true. The exterior angle and the adjacent interior angle form a linear pair (sum 180 degrees), so R is false.
44	**(c)** A is true but R is false.	A is true. Dividing by a fraction means multiplying by its reciprocal, so R is false.
45	**(d)** A is false but R is true.	10 lakh = 10,00,000 has 6 zeros, so A is false. R is true.
46	**(d)** A is false but R is true.	12 / 3 x 2 = 4 x 2 = 8, so A is false. R is true.
47	**(d)** A is false but R is true.	3 x 2 = 6 with two decimal places gives 0.06, so A is false. R is true.
48	**(d)** A is false but R is true.	4x and 3y are unlike terms and cannot be combined, so A is false. R is true.
49	**(d)** A is false but R is true.	Corresponding angles are equal, so A is false and R is true.
50	**(d)** A is false but R is true.	The product of two odd numbers is odd (e.g. 3 x 5 = 15), so A is false. R is true.

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